

AUG
2013

 **Fenner Drives®**



EAGLE®
POLYURETHANE BELTING & O-RINGS

Non-Reinforced Polyurethane Belting

Eagle Blue 80 EC
Eagle Clear 80 EC
Eagle Blue 80 MD
Eagle Brown 80
Eagle Opaque 80
Eagle Blue 85
Eagle Clear 85
Eagle Ivory 85
Eagle Orange 85
Eagle Red 85
Eagle Green 89
Eagle Green 89 T
Eagle Red 90
Eagle Beige 95
Eagle Clear 95



Eagle® Non-Reinforced Polyurethane Belting — the proven workhorses for material transfer and light-duty power transmission applications.

- Solid polyurethane construction
- Round, V- and flat profiles
- Excellent abrasion resistance
- Self tensioning — no take-ups required
- Easily welded on site with a Fenner Drives Welding Kit

Part Number Listing

Round Profiles – Non-Reinforced

	Eagle Blue 80 EC	Eagle Clear 80 EC	Eagle Blue 80 MD	Eagle Brown 80	Eagle Opaque 80	Eagle Blue 85	Eagle Clear 85	Eagle Orange 85	Eagle Red 85
3/32"							4908003	1032003	
1/8"							4908006	1032004	
3/16"			4941100				4908009	1032006	
1/4"			4941101		4940003		4908012	1032008	
5/16"			4941102				4908015	1032010	
3/8"			4941103		4940005		4908018	1032012	
1/2"			4941105		4940011		4908024	1032016	
9/16"			4941106				4908027	1032018	
5/8"			4941107				4908030	1032020	
3/4"							4908033	1032024	
2mm	4928000	4927000		L04BR802M	L04OP802M		L04C852M	L04OG852M	
3mm	4928001	4927001		L04BR803M	L04OP803M	L04BL853M	L04C853M	L04OG853M	L04R853M
4mm	4928002	4927002		L04BR804	L04OP804	L04BL854	L04C854	L04OG854	L04R8504
5mm	4928003	4927003		L04BR805M	L04OP805M	L04BL855M	L04C855M	4940100	L04R855
6mm	4928004	4927004			L04OP806M	L04BL856M	L04C856M	L04OG856M	
6.3mm	4928005	4927005		L04BR806.3		L04BL856.3			L04R856.3
7mm					L04OP807M		L04C857M	L04OG857M	
8mm	4928006	4927006		L04BR808	L04OP808M	L04BL858M	L04C858	L04OG858M	L04R858
9.5mm	4928007	4927007		L04BR809.5M		L04BL859.5M			L04R859.5
10mm	4928008	4927008			L04OP8010M	L04BL8510M	L04C8510M	L04OG8510M	
12mm							L04C8512M	L04OG8512M	
12.7mm				L04BR80127		L04BL8512.7			L04R8512.7
15mm				L04BR8015M	L04OP8015M	L04BL8515			L04R8515
18mm				L04BR8018M	4940013				

	Eagle Green 89	Eagle Green 89 T	Eagle Red 90	Eagle Beige 95	Eagle Clear 95	Eagle White 40D	Eagle Blue 55D
3/32"					4907003		
1/8"					4907006		
3/16"			4940022		4907009		
1/4"			4940023		4907012		
5/16"					4907015		
3/8"			4940025		4907018		
1/2"			4940026		4907024		
9/16"			4940036		4907027		
5/8"					4907030		
3/4"					4907033		
2mm	4905402	4905302	4940017				
3mm	L04G893MS	4905303	4940020			L04BY403M	
4mm	4905404	4905304	4940021			L04BY404	
5mm	4905405	4905305	L04R9005M	L04BE955M		L04BY405M	
6mm	4905406	4905306				L04BY406M	
6.3mm	L04896.3MS						
7mm	4999637	4905307	L04R907				
8mm	L04G898MS	4905308	4940024	L04BE958		L04BY408	
9mm	L04G899MS	L04G899					
9.5mm	L04G899.5MS						
10mm	L04G8910MS	4905310	L04R9010M	L04BE9510M		L04BY4010M	L04BY5510M
12mm	L04G8912MS	4905312	L04R9012M			L04BY4012	
15mm	L04G8915MS	4905315	4999315	L04BE9515M		L04BY4015	L04BY5515
18mm	L04G8918MS	4940091				L04BY4018	L04BY5518
20mm	L04G8920MS	4940092				L04BY4020	

All belting sold in 100'/30.5m lengths, except Can Cable (sold in 500' lengths).

May be subject to minimum order. Consult factory for availability. Dimensions are for reference only.

For technical assistance and drive design help, contact Applications Engineering at 800-243-3374 or ae@fennerdrives.com.

Eagle Blue 80 EC				HARDNESS 80A FDA COMPLIANT Yes		COEFFICIENT OF FRICTION Stainless Steel .80 Steel .70 UHMW .55				TEMPERATURE RANGE -22°F to +150°F -30°C to +66°C					
DESCRIPTION Round, Non-Reinforced															
Cross Section	Part Number	Dimensions Ø (in) (mm)		Minimum Pulley Ø (in) (mm)		Working Load @ Percent Tension								Wt/ft (lbs)	Wt/m (kg)
						4%		6%		8%		10%			
2mm	4928000	2	0.55 14	0.1 0.7	0.2 1.0	0.3 1.4	0.4 1.7	0.002	0.004						
3mm	4928001	3	0.83 21	0.4 1.7	0.6 2.5	0.8 3.5	1.0 4.3	0.005	0.008						
4mm	4928002	4	1.10 28	0.6 2.8	0.9 4.2	1.3 5.6	1.5 6.7	0.009	0.014						
5mm	4928003	5	1.38 35	0.9 4.2	1.4 6.3	1.9 8.4	2.4 10.6	0.015	0.022						
6mm	4928004	6	1.65 42	1.3 5.9	2.1 9.1	2.8 12.3	3.4 15.2	0.021	0.032						
6.3mm	4928005	1/4 6.3	1.74 44	1.5 6.6	2.3 10.2	3.1 13.7	3.8 17.0	0.023	0.035						
8mm	4928006	8	2.20 56	2.4 10.5	3.6 16.2	4.8 21.5	6.1 26.9	0.038	0.056						
9.5mm	4928007	3/8 9.5	2.62 67	3.4 15.0	5.2 23.2	7.0 31.0	8.7 38.7	0.053	0.079						
10mm	4928008	10	2.76 70	3.7 16.4	5.8 25.6	7.7 34.1	9.6 42.6	0.059	0.088						

Eagle Blue 80 EC DESCRIPTION Trapezoidal, Non-Reinforced			HARDNESS 80A FDA COMPLIANT Yes		COEFFICIENT OF FRICTION Stainless Steel .80 Steel .70 UHMW .55				TEMPERATURE RANGE -22°F to +150°F -30°C to +66°C					
Vee T-Top														
Cross Section	Part Number	Dimensions w × h* (mm)	Minimum Pulley Ø (in) (mm)		Working Load @ Percent Tension								Wt/ft (lbs)	Wt/m (kg)
					4% (lbs) (N)		6% (lbs) (N)		8% (lbs) (N)		10% (lbs) (N)			
6mm × 4mm	4928009	6.5 × 4	1.10	28	0.8	3.7	1.5	6.7	2.1	9.4	2.7	12.1	0.015	0.023
8mm × 5mm	4928010	8 × 5	1.38	35	1.3	5.7	2.4	10.5	3.3	14.6	4.2	18.9	0.024	0.035
10mm × 4mm T-Top	4928011	10 × 4	1.10	28	1.0	4.6	1.9	8.5	2.7	11.8	3.4	15.3	0.019	0.028
Z/10	4928012	10 × 6.5	1.79	46	2.1	9.2	3.8	16.8	5.3	23.5	6.8	30.3	0.038	0.057
A/13	4928013	13 × 8	2.20	56	3.4	14.9	6.1	27.3	8.6	38.2	11.1	49.3	0.062	0.092
B/17	4928014	17 × 11.5	3.17	81	6.1	27.3	11.2	49.9	15.7	69.9	20.3	90.2	0.113	0.168

Eagle Clear 80 EC				HARDNESS 80A FDA COMPLIANT Yes		COEFFICIENT OF FRICTION Stainless Steel .80 Steel .70 UHMW .55				TEMPERATURE RANGE -22°F to +150°F -30°C to +66°C					
DESCRIPTION Round, Non-Reinforced															
Cross Section	Part Number	Dimensions Ø (in) (mm)		Minimum Pulley Ø (in) (mm)		Working Load @ Percent Tension								Wt/ft (lbs)	Wt/m (kg)
						4%		6%		8%		10%			
2mm	4927000		2	0.55	14	0.1	0.7	0.2	1.0	0.3	1.4	0.4	1.7	0.002	0.004
3mm	4927001		3	0.83	21	0.4	1.7	0.6	2.5	0.8	3.5	1.0	4.3	0.005	0.008
4mm	4927002		4	1.10	28	0.6	2.8	0.9	4.2	1.3	5.6	1.5	6.7	0.009	0.014
5mm	4927003		5	1.38	35	0.9	4.2	1.4	6.3	1.9	8.4	2.4	10.6	0.015	0.022
6mm	4927004		6	1.65	42	1.3	5.9	2.1	9.1	2.8	12.3	3.4	15.2	0.021	0.032
6.3mm	4927005	¼	6.3	1.74	44	1.5	6.6	2.3	10.2	3.1	13.7	3.8	17.0	0.023	0.035
8mm	4927006		8	2.20	56	2.4	10.5	3.6	16.2	4.8	21.5	6.1	26.9	0.038	0.056
9.5mm	4927007	⅜	9.5	2.62	67	3.4	15.0	5.2	23.2	7.0	31.0	8.7	38.7	0.053	0.079
10mm	4927008		10	2.76	70	3.7	16.4	5.8	25.6	7.7	34.1	9.6	42.6	0.059	0.088

Eagle Clear 80 EC			HARDNESS 80A		COEFFICIENT OF FRICTION Stainless Steel .80 Steel .70 UHMW .55				TEMPERATURE RANGE -22°F to +150°F -30°C to +66°C					
DESCRIPTION Trapezoidal, Non-Reinforced			FDA COMPLIANT Yes											
Cross Section	Part Number	Dimensions w × h* (mm)	Minimum Pulley Ø (in) (mm)		Working Load @ Percent Tension								Wt/ft (lbs)	Wt/m (kg)
					4% (lbs) (N)	6% (lbs) (N)	8% (lbs) (N)	10% (lbs) (N)						
6mm × 4mm	4927009	6.5 × 4	1.10	28	0.8	3.7	1.5	6.7	2.1	9.4	2.7	12.1	0.015	0.023
8mm × 5mm	4927010	8 × 5	1.38	35	1.3	5.7	2.4	10.5	3.3	14.6	4.2	18.9	0.024	0.035
10mm × 4mm T-Top	4927011	10 × 4	1.10	28	1.0	4.6	1.9	8.5	2.7	11.8	3.4	15.3	0.019	0.028
Z/10	4927012	10 × 6.5	1.79	46	2.1	9.2	3.8	16.8	5.3	23.5	6.8	30.3	0.038	0.057
A/13	4927013	13 × 8	2.20	56	3.4	14.9	6.1	27.3	8.6	38.2	11.1	49.3	0.062	0.092
B/17	4927014	17 × 11.5	3.17	81	6.1	27.3	11.2	49.9	15.7	69.9	20.3	90.2	0.113	0.168

For technical assistance and drive design help, contact Applications Engineering at ae@fennerdrives.com.

* w (width) is the widest part of the belt. h (height) is the tallest part of the belt, NOT including the belting top surface.
Dimensions are for reference only.

Engineering Data — Metric Pulley Sections

V-Belts

Eagle® V-belts in “classical” Z/10, A/13, B/17, C/22 and D/32 cross sections are designed to fit ISO and DIN 2215 compliant pulleys as per the groove details illustrated in Figure 1.

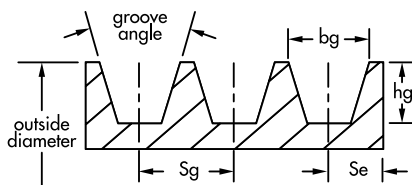


Figure 1

Cross Section	Datum Diameter Range	Groove Angle	b_g (mm)	h_g Min (mm)	S_g (mm)	S_e (mm)
Z/10	Up thru 80mm Over 80mm	$34^\circ \pm 1^\circ$ $38^\circ \pm 1^\circ$	9.7	11	12 ± 0.3	8 ± 0.6
A/13	Up thru 118mm Over 118mm	$34^\circ \pm 1^\circ$ $38^\circ \pm 1^\circ$	12.7	14	15 ± 0.3	10 ± 0.6
B/17	Up thru 190mm Over 190mm	$34^\circ \pm 1^\circ$ $38^\circ \pm 1^\circ$	16.3	18	19 ± 0.4	12.5 ± 0.8
C/22	Up thru 315mm Over 315mm	$34^\circ \pm 1^\circ$ $38^\circ \pm 30'$	22	24	25.5 ± 0.5	17 ± 1.0
D/32	Up thru 500mm Over 500mm	$36^\circ \pm 30'$ $38^\circ \pm 30'$	32	28	37 ± 0.6	24 ± 2.0

Dimensions in millimetres unless otherwise indicated.

Round Belts

Eagle round belts are commonly run in pulleys with a round groove; see Figure 2a. In the absence of round groove pulleys, they can also be used in V-groove pulleys (Figure 2b). The table at right shows the dimensional data for a round belt used in a V-groove pulley.

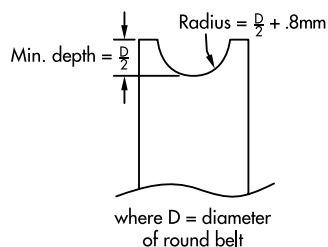


Figure 2a

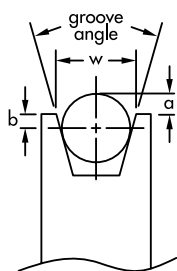


Figure 2b

Pulley Size	Pulley Diameter O.D. (mm)	Groove Angle	Round Belt	Dimensions		
				w	a	b
Z/10	Up thru 80mm	34°	7	9.7	-0.39	3.89
			8	9.7	1.82	2.18
			9.5	9.7	5.14	-0.39
Z/10	Over 80mm	38°	7	9.7	0.17	3.34
			8	9.7	2.19	1.81
			9.5	9.7	5.25	-0.50
A/13	Up thru 118mm	34°	9.5	12.7	0.23	4.52
			10	12.7	1.33	3.67
			12	12.7	5.75	0.25
A/13	Over 118mm	38°	9.5	12.7	0.90	3.85
			10	12.7	1.91	3.09
			12	12.7	5.98	0.02
B/17	Up thru 190mm	34°	12	16.3	-0.14	6.14
			15	16.3	6.50	1.00
			16	16.3	8.71	-0.71
B/17	Over 190mm	38°	12	16.3	0.76	5.24
			15	16.3	6.87	0.63
			16	16.3	8.90	-0.90
C/22	Up thru 315mm	34°	20	22	8.22	1.78
C/22	Over 315mm	38°	20	22	9.00	1.23

Note: above dimensions are belt fit in groove under no tension.
Dimensions are in millimetres unless otherwise indicated.

Flat Belts

All flat belts have a natural tendency to move laterally. Therefore a flat or straight pulley is not recommended, as the belt would walk off the pulley. To keep the belt in the centre of the pulley it must have a crown. Figure 3a illustrates a round crown and is the preferred method. A modified round crown as illustrated in Figure 3b is also acceptable. A flat pulley with guide flanges (Figure 3c) is not recommended. Even with the guide flanges the belt will move laterally and potentially could climb up onto them.

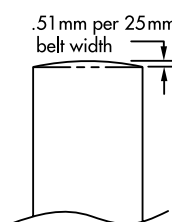


Figure 3a

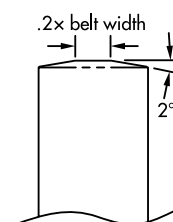


Figure 3b

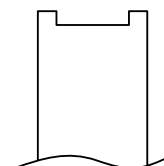


Figure 3c

Belt Installation Tension

All belts require a certain amount of tension to function properly in the application. The specific installation tension is determined from several factors including belt type, construction and working load. Belt details are in the Technical Data section of this catalog and working load is derived from your application.

Non-Reinforced Belting: When non-reinforced belting is stretched and released, elasticity is the property that brings the material back to its original shape. This “memory” is what gives our non-reinforced belting its self-tensioning properties. When a non-reinforced belt is first installed (stretched) the material does not return to 100% of its original length and continues to lose elasticity over its life span. This loss in elasticity is evident as tension decay. To overcome tension decay effects, a non-reinforced belt requires a relatively high install tension. Installation tensions ranging from 6% to 10% will normally be sufficient for most applications. If higher tensions are required, the application may exceed the belt’s load capacity.

Reinforced Belting: Reinforced belts contain a reinforcing tensile member which increases the belt’s modulus of elasticity. This reduces the belt’s ability to stretch and minimizes tension decay. This allows a reinforced belt to carry a greater load than a non-reinforced belt. Since an endless reinforced belt is essentially a fixed length, it cannot be stretched on like a non-reinforced belt. Consequently, reinforced belts require a mechanical take-up mechanism to apply the appropriate installation tension as well as accommodating any eventual small amount of tension decay that may occur. This mechanism should accommodate at least 4% of the belt’s length.

Belt Installation Length

In this section, we will refer to two different lengths that are defined as follows:

1. Reference Length: The length determined by taking a measuring tape and following the path of the belt around all of the pulleys, or through computer aided design (CAD) techniques. This length may also be obtained from the equation below. Take up mechanisms should be adjusted to the minimum position to allow for maximum adjustment of the belt prior to taking or calculating length. Note: this equation applies to two-pulley drives only.

$$L = 2C + \frac{\pi}{2}(D + d) + \frac{(D - d)^2}{4C}$$

where: L = reference length
C = centre of pulley shaft to centre of pulley shaft distance
D = pitch diameter of large pulley
d = pitch diameter of small pulley

2. Cut Length: The length the belt is cut to prior to welding.

Apply the following formulas to determine the Cut Length from Reference Length:

Butt weld non-reinforced:

Cut Length = Reference Length ÷ (1 + % tension)

Example: Reference Length for a non-reinforced belt is 44" (1120mm), requires 8% tension and will be butt welded. Cut Length is calculated on right.

Cut Length = 44" ÷ (1 + 8%)	Cut Length = 1120mm ÷ (1 + 8%)
= 44" ÷ 1.08	= 1120mm ÷ 1.08
= 40.7"	= 1037mm

Overlap weld reinforced: *Cut Length = Reference Length + 1.5" (38mm)*

Example: Reference Length for a reinforced belt is 44" (1120mm) and will be overlap welded. The overlap weld consumes 1.5" (38mm) of belt length. Cut Length is calculated on right.

Cut Length = 44" + 1.5"	Cut Length = 1120mm + 38mm
= 45.5"	= 1158mm

Butt weld reinforced: *Cut Length = Reference Length*

Example: Reference Length for a reinforced belt is 44" (1120mm) and will be butt welded. The weld consumes a negligible amount of belt length, consequently, Cut Length and Reference Length are the same. Cut Length is calculated on right.

Cut Length = 44"	Cut Length = 1120mm
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Temperature

The temperature range of polyurethane belting is determined by the thermoplastic resin. Like all thermoplastic resins its physical properties change with changes in temperature. At higher temperatures the material will soften, lose strength and can elongate excessively to the point of premature failure. At lower temperatures the material will become more brittle and stiff which can result in cracking. The temperature ranges are for guidance and listed under each individual belt type in the Technical Data section.

Minimum Pulley Diameter

The most common serious mistake in designing belt drives is the selection of a pulley diameter that is too small. In most cases, non-reinforced belts can operate on smaller diameter pulleys than belts with a reinforcing tensile member. Reinforced belts require a larger pulley diameter to prevent premature flex fatigue failure of the tensile member. Listed under each individual belt type in the Technical Data section is the recommended minimum pulley diameter. Smaller diameters can be used only if a reduction in belt service life is acceptable.

Belt Profile Tolerances

Round Belts:

Up to and including 3/16" (5mm) diameter:	± .005" (± 0.13mm)
Over 3/16" (5mm) up to and including 1/4" (7mm) diameter:	± .007" (± 0.18mm)
Over 1/4" (7mm) up to and including 9/16" (15mm) diameter:	± .010" (± 0.25mm)
Over 9/16" (15mm) up to and including 5/8" (18mm) diameter:	± .012" (± 0.30mm)
Over 5/8" (18mm) diameter:	± .015" (± 0.38mm)

Flat and V-Belts:

All profiles: ± .015" (± 0.38mm)

If tighter tolerances are required, consult Fenner Drives Applications Engineering Group with your requirements.

Engineering Data — Selection Procedure, Conveying

1. Refer to the Technical Data chart for the belt material and cross section selected.
2. Use the following formula that meets your application requirements (Note: if belt supported by rollers use .17 for μ):
 - a. Horizontal Transport with Slider Bed

$$T_e = W_t \times \mu + B_{wt}$$
 - b. Horizontal Transport with Slider Bed and Product Accumulation

$$T_e = W_t \times \mu + B_{wt} + A_{wt}$$
 - c. Incline or Decline Transport with Slider Bed

$$T_e = \frac{W_t}{C} (H_t + \mu \times \sqrt{C^2 + H_t^2}) + B_{wt}$$
 - d. Incline or Decline Transport with Slider Bed and Product Accumulation

$$T_e = \frac{W_t}{C} (H_t + \mu \times \sqrt{C^2 + H_t^2}) + B_{wt} + A_{wt}$$
3. Determine Tight Tension (T_1).
 Flat and round belts: $T_1 = T_e \times 2$
 V-belts: $T_1 = T_e \times 1.25$
4. Refer to the Technical Data chart for the material and cross section selected and compare T_1 to the Working Load at maximum % tension. If only one belt is desired, T_1 may not be greater than the Working Load at maximum % tension. If more than one belt is required, divide T_1 by the Working Load at maximum % tension to arrive at number of belts. Round up to the nearest whole number of belts.
5. Find load per belt by dividing T_1 by number of belts. From the Technical Data chart, determine the percent installed tension for the load per belt.

Where:

T_e = Effective Tension

W_t = Total Weight on Conveyor

C = Conveyor Centre Distance

B_{wt} = Belt weight/unit length $\times C$

A_{wt} = Accumulating weight $\times \mu'$
 (where μ' is the COF between belt and product)

H_t = Incline or decline height

μ = COF on slider bed material from chart

To determine the required belt length, please refer to the "Belt Installation Length" section on the previous page.

Engineering Data — Selection Example

Eagle Orange 85

DESCRIPTION
Round,
Non-Reinforced



HARDNESS

85A

FDA COMPLIANT
Yes

COEFFICIENT OF FRICTION

Stainless Steel .70

Steel .60

UHMW .45

TEMPERATURE RANGE

-22°F to +150°F

-30°C to +66°C

Cross Section	Part Number	Dimensions Ø (in) (mm)		Minimum Pulley Ø (in) (mm)		Working Load @ Percent Tension								Wt/ft (lbs)	Wt/m (kg)
						4% (lbs) (N)		6% (lbs) (N)		8% (lbs) (N)		10% (lbs) (N)			
6mm	L04OG856M	6		1.89	48	1.7	7.7	2.7	11.8	3.5	15.8	4.4	19.4	0.023	0.034
¼"	1032008	¼	6.3	2.00	51	1.9	8.6	3.0	13.3	4.0	17.7	4.9	21.9	0.026	0.038

Example 1

Type of belt being considered = Eagle Orange 85 in 1/4" round

Head-to-tail centre distance (C) = 10 feet

Incline or decline = none

Product accumulation on belt(s)? = no

Total weight on belt(s) = 15 lbs.

Type of belt support = UHMW slider bed

2. Horizontal Transport with Slider Bed.
 Since the belt will run in UHMW slider bed the COF(μ) of .45 is used from Technical Data chart. From the chart the belt weight is .026 lbs/ft giving a total belt weight of .26 lbs (.026 $\times$ 10').
 $T_e = 15 \text{ lbs} \times .45 + .26 = 7.01$
3. Determine Tight Tension (T_1).
 round belts $T_1 = 7.01 \times 2 = 14.02$
4. Refer to the Technical Data chart for the material and cross section selected and compare T_1 to the Working Load at 10% tension. If only one belt is desired, T_1 may not be greater than the Working Load at 10% tension. If more than one belt is required, divide T_1 by the Working Load at 10% tension to arrive at number of belts. Round up to the nearest whole number of belts.
 1/4" round rated 4.9 lbs @ 10% tension. $14.02 \div 4.9 = 2.86$ use 3 belts
5. Find load per belt by dividing T_1 by number of belts. From the Technical Data chart, determine the percent installed tension for the load per belt.
 Load/belt = $14.02 \div 3 = 4.67$ lbs
 corresponding installed tension = 9.7%

Example 2

Eagle Orange 85 in 6mm round

Head-to-tail centre distance (C) = 3 metres

Incline or decline = none

Product accumulation on belt(s)? = no

Total weight on belt(s) = 6 kg

Type of belt support = UHMW slider bed

2. Horizontal Transport with Slider Bed.
 Since the belt will run in UHMW slider bed the COF(μ) of .45 is used from Technical Data chart. From the chart the belt weight is .034 kgs/m giving a total belt weight of .102 kg (.034 $\times$ 3M).
 $T_e = 6 \text{ kg} \times .45 + .102 = 2.802 \text{ kg}$
3. Determine Tight Tension (T_1).
 round belts $T_1 = 2.802 \times 2 = 5.604 \text{ kg} = 54.98 \text{ Newtons} (5.604 \times 9.81)$
4. Refer to the Technical Data chart for the material and cross section selected and compare T_1 to the Working Load at 10% tension. If only one belt is desired, T_1 may not be greater than the Working Load at 10% tension. If more than one belt is required, divide T_1 by the Working Load at 10% tension to arrive at number of belts. Round up to the nearest whole number of belts.
 6mm round rated 19.4 kg @ 10% tension. $54.98 \div 19.4 = 2.83$ use 3 belts
5. Find load per belt by dividing T_1 by number of belts. From the Technical Data chart, determine the percent installed tension for the load per belt.
 Load/belt = $54.98 \text{ N} \div 3 = 18.33 \text{ Newtons}$
 corresponding installed tension = 9.6%